Chapter 1

# Introduction

## 1.1 Before You Start

Statistics starts with a problem, proceeds with the collection of data, continues with the data analysis and finishes with conclusions. It is a common mistake of inexperienced statisticians to plunge into a complex analysis without paying attention to the objectives or even whether the data are appropriate for the proposed analysis. As Einstein said, the formulation of a problem is often more essential than its solution which may be merely a matter of mathematical or experimental skill.

To formulate the problem correctly, you must:

1. Understand the physical background. Statisticians often work in collaboration with others and need to understand something about the subject area. Regard this as an opportunity to learn something new rather than a chore.
2. Understand the objective. Again, often you will be working with a collaborator who may not be clear about what the objectives are. Beware of "fishing expeditions" — if you look hard enough, you will almost always find something, but that something may just be a coincidence.
3. Make sure you know what the client wants. You can often do quite different analyses on the same dataset. Sometimes statisticians perform an analysis far more complicated than the client really needed. You may find that simple descriptive statistics are all that are needed.
4. Put the problem into statistical terms. This is a challenging step and where irreparable errors are sometimes made. Once the problem is translated into the language of statistics, the solution is often routine. Difficulties with this step explain why artificial intelligence techniques have yet to make much impact in application to statistics. Defining the problem is hard to program. That a statistical method can read in and process the data is not enough. The results of an inapt analysis may be meaningless.

It is important to understand how the data were collected.

1. Are the data observational or experimental? Are the data a sample of convenience or were they obtained via a designed sample survey. How the data were collected has a crucial impact on what conclusions can be made.
2. Is there nonresponse? The data you do not see may be just as important as the data you do see.
3. Are there missing values? This is a common problem that is troublesome and time consuming to handle.

4. How are the data coded? In particular, how are the categorical variables represented?
5. What are the units of measurement?
6. Beware of data entry errors and other corruption of the data. This problem is all too common — almost a certainty in any real dataset of at least moderate size. Perform some data sanity checks.

## 1.2 Initial Data Analysis

This is a critical step that should always be performed. It is simple but it is vital. You should make numerical summaries such as means, standard deviations (SDs), maximum and minimum, correlations and whatever else is appropriate to the specific dataset. Equally important are graphical summaries. There is a wide variety of techniques to choose from. For one variable at a time, you can make boxplots, histograms, density plots and more. For two variables, scatterplots are standard while for even more variables, there are numerous good ideas for display including interactive and dynamic graphics. In the plots, look for outliers, data-entry errors, skewed or unusual distributions and structure. Check whether the data are distributed according to prior expectations.

Getting data into a form suitable for analysis by cleaning out mistakes and aberrations is often time consuming. It often takes more time than the data analysis itself. In this book, all the data will be ready to analyze, but you should realize that in practice this is rarely the case.

Let's look at an example. The National Institute of Diabetes and Digestive and Kidney Diseases conducted a study on 768 adult female Pima Indians living near Phoenix. The following variables were recorded: number of times pregnant, plasma glucose concentration at 2 hours in an oral glucose tolerance test, diastolic blood pressure (mmHg), triceps skin fold thickness (mm), 2-hour serum insulin (mu U/ml), body mass index (weight in kg/(height in $m^2$)), diabetes pedigree function, age (years) and a test whether the patient showed signs of diabetes (coded zero if negative, one if positive). The data may be obtained from UCI Repository of machine learning databases at `archive.ics.uci.edu/ml`.

Before doing anything else, one should find out the purpose of the study and more about how the data were collected. However, let's skip ahead to a look at the data:

```
> data(pima, package="faraway")
> head(pima)
  pregnant glucose diastolic triceps insulin  bmi diabetes age test
1        6     148         72       35       0 33.6    0.627  50    1
2        1      85         66       29       0 26.6    0.351  31    0
3        8     183         64        0       0 23.3    0.672  32    1
4        1      89         66       23      94 28.1    0.167  21    0
5        0     137         40       35     168 43.1    2.288  33    1
6        5     116         74        0       0 25.6    0.201  30    0
```

The `data(pima, package="faraway")` makes the data available from the `faraway` package. An alternative is to just type `require(faraway)` which makes all the data

and functions specific to this book available. You do need to install this package first as explained in Appendix A. If you get an error message about data not being found, it may be that you have forgotten to install the package.

The command `head(pima)` prints out the first six lines of the data frame. This is a good way to see what variables we have and what sort of values they take. You can type `pima` to see the whole data frame but 768 lines may be more than you want to examine.

We start with some numerical summaries:

```
> summary(pima)
    pregnant         glucose       diastolic        triceps
 Min.   : 0.00   Min.   :  0   Min.   :  0.0   Min.   : 0.0
 1st Qu.: 1.00   1st Qu.: 99   1st Qu.: 62.0   1st Qu.: 0.0
 Median : 3.00   Median :117   Median : 72.0   Median :23.0
 Mean   : 3.85   Mean   :121   Mean   : 69.1   Mean   :20.5
 3rd Qu.: 6.00   3rd Qu.:140   3rd Qu.: 80.0   3rd Qu.:32.0
 Max.   :17.00   Max.   :199   Max.   :122.0   Max.   :99.0
    insulin           bmi          diabetes          age
 Min.   :  0.0   Min.   : 0.0   Min.   :0.078   Min.   :21.0
 1st Qu.:  0.0   1st Qu.:27.3   1st Qu.:0.244   1st Qu.:24.0
 Median : 30.5   Median :32.0   Median :0.372   Median :29.0
 Mean   : 79.8   Mean   :32.0   Mean   :0.472   Mean   :33.2
 3rd Qu.:127.2   3rd Qu.:36.6   3rd Qu.:0.626   3rd Qu.:41.0
 Max.   :846.0   Max.   :67.1   Max.   :2.420   Max.   :81.0
      test
 Min.   :0.000
 1st Qu.:0.000
 Median :0.000
 Mean   :0.349
 3rd Qu.:1.000
 Max.   :1.000
```

The `summary()` command is a quick way to get the usual univariate summary information. At this stage, we are looking for anything unusual or unexpected, perhaps indicating a data-entry error. For this purpose, a close look at the minimum and maximum values of each variable is worthwhile. Starting with `pregnant`, we see a maximum value of 17. This is large, but not impossible. However, we then see that the next five variables have minimum values of zero. No blood pressure is not good for the health — something must be wrong. Let's look at the sorted values:

```
> sort(pima$diastolic)
  [1]   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
 [17]   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
 [33]   0   0   0  24  30  30  38  40  44  44  44  44  46  46  48  48
...etc...
```

We see that the first 35 values are zero. The description that comes with the data says nothing about it but it seems likely that the zero has been used as a missing value code. For one reason or another, the researchers did not obtain the blood pressures of 35 patients. In a real investigation, one would likely be able to question the researchers about what really happened. Nevertheless, this does illustrate the kind of misunderstanding that can easily occur. A careless statistician might overlook these presumed missing values and complete an analysis assuming that these were real ob-

served zeros. If the error was later discovered, they might then blame the researchers for using zero as a missing value code (not a good choice since it is a valid value for some of the variables) and not mentioning it in their data description. Unfortunately such oversights are not uncommon, particularly with datasets of any size or complexity. The statistician bears some share of responsibility for spotting these mistakes.

We set all zero values of the five variables to NA which is the missing value code used by R:

```
> pima$diastolic[pima$diastolic == 0]  <- NA
> pima$glucose[pima$glucose == 0] <- NA
> pima$triceps[pima$triceps == 0]  <- NA
> pima$insulin[pima$insulin == 0] <- NA
> pima$bmi[pima$bmi == 0] <- NA
```

The variable test is not quantitative but categorical. Such variables are also called *factors*. However, because of the numerical coding, this variable has been treated as if it were quantitative. It is best to designate such variables as factors so that they are treated appropriately. Sometimes people forget this and compute stupid statistics such as the "average zip code."

```
> pima$test <- factor(pima$test)
> summary(pima$test)
  0   1
500 268
```

We now see that 500 cases were negative and 268 were positive. It is even better to use descriptive labels:

```
> levels(pima$test) <- c("negative","positive")
> summary(pima)
    pregnant         glucose         diastolic         triceps
 Min.   : 0.000   Min.   : 44.0   Min.   : 24.0   Min.   :  7.00
 1st Qu.: 1.000   1st Qu.: 99.0   1st Qu.: 64.0   1st Qu.: 22.00
 Median : 3.000   Median :117.0   Median : 72.0   Median : 29.00
 Mean   : 3.845   Mean   :121.7   Mean   : 72.4   Mean   : 29.15
 3rd Qu.: 6.000   3rd Qu.:141.0   3rd Qu.: 80.0   3rd Qu.: 36.00
 Max.   :17.000   Max.   :199.0   Max.   :122.0   Max.   : 99.00
                  NA's   :  5.0   NA's   : 35.0   NA's   :227.00
    insulin            bmi           diabetes           age
 Min.   : 14.00   Min.   :18.20   Min.   :0.0780   Min.   :21.00
 1st Qu.: 76.25   1st Qu.:27.50   1st Qu.:0.2437   1st Qu.:24.00
 Median :125.00   Median :32.30   Median :0.3725   Median :29.00
 Mean   :155.55   Mean   :32.46   Mean   :0.4719   Mean   :33.24
 3rd Qu.:190.00   3rd Qu.:36.60   3rd Qu.:0.6262   3rd Qu.:41.00
 Max.   :846.00   Max.   :67.10   Max.   :2.4200   Max.   :81.00
 NA's   :374.00   NA's   :11.00
       test
 negative:500
 positive:268
```

Now that we have cleared up the missing values and coded the data appropriately, we are ready to do some plots. Perhaps the most well-known univariate plot is the histogram:

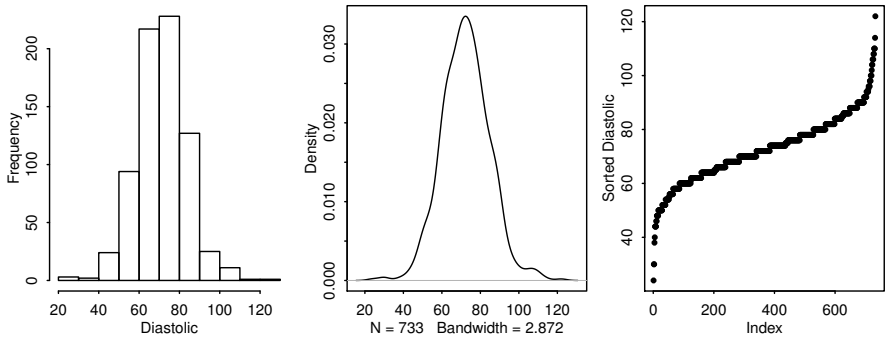



Figure 1.1 The first panel shows a histogram of the diastolic blood pressures, the second shows a kernel density estimate of the same, while the third shows an index plot of the sorted values.

```
> hist(pima$diastolic,xlab="Diastolic",main="")
```

as seen in the first panel of Figure 1.1. We see a bell-shaped distribution for the diastolic blood pressures centered around 70. The construction of a histogram requires the specification of the number of bins and their spacing on the horizontal axis. Some choices can lead to histograms that obscure some features of the data. R specifies the number and spacing of bins given the size and distribution of the data, but this choice is not foolproof and misleading histograms are possible. For this reason, some prefer to use kernel density estimates, which are essentially a smoothed version of the histogram (see Simonoff (1996) for a discussion of the relative merits of histograms and kernel estimates):

```
> plot(density(pima$diastolic,na.rm=TRUE),main="")
```

The kernel estimate may be seen in the second panel of Figure 1.1. We see that this plot avoids the distracting blockiness of the histogram. Another alternative is to simply plot the sorted data against its index:

```
> plot(sort(pima$diastolic),ylab="Sorted Diastolic")
```

The advantage of this is that we can see all the cases individually. We can see the distribution and possible outliers. We can also see the discreteness in the measurement of blood pressure — values are rounded to the nearest even number and hence we see the "steps" in the plot.

Now we show a couple of bivariate plots, as seen in Figure 1.2:

```
> plot(diabetes ~ diastolic,pima)
> plot(diabetes ~ test,pima)
```

First, we see the standard scatterplot showing two quantitative variables. Second, we see a side-by-side boxplot suitable for showing a quantitative with a qualitative variable.

The base graphics system in R is fine for producing standard plots. A useful

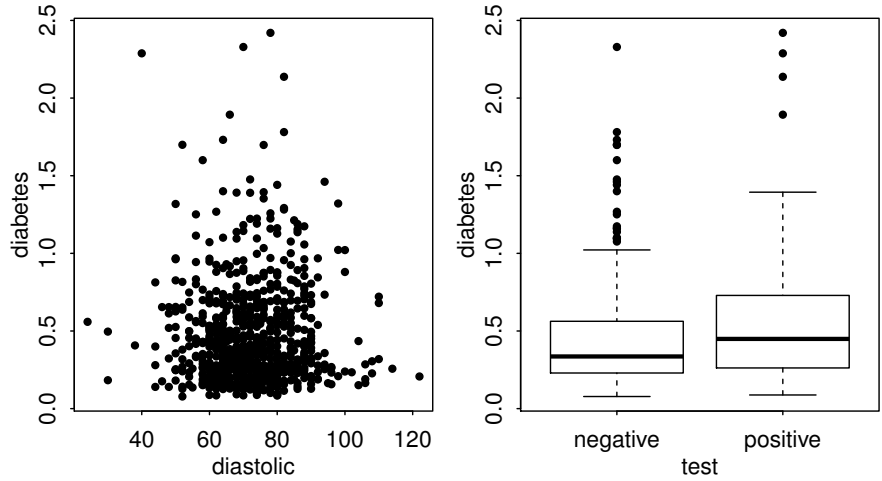



Figure 1.2 The first panel shows scatterplot of the diastolic blood pressures against diabetes function and the second shows boxplots of diastolic blood pressure broken down by test result.

alternative is provided by the `ggplot2` package which you will need to install as described in Appendix A. Here's how we can produce versions of the same plots:

```
> require(ggplot2)
> ggplot(pima,aes(x=diastolic))+geom_histogram()
> ggplot(pima,aes(x=diastolic))+geom_density()
> ggplot(pima,aes(x=diastolic,y=diabetes))+geom_point()
```

In each case, the first part of the command specifies where the data comes from (the data frame is `pima`) and an *aesthetic* using `aes`. The aesthetic specifies what you see such as position in the $x$ or $y$ direction or aspects such as shape or color. The second part of the command in each case is specifying the particular geometry for the plot. The results are shown in Figure 1.3. The relative advantage of `ggplot2` is more apparent in producing more complex plots involving more than just two variables. For example, we show two different ways that the varying `test` can be shown in the relationship between `diastolic` and `diabetes`:

```
> ggplot(pima,aes(x=diastolic,y=diabetes,shape=test))+geom_point()+
    theme(legend.position = "top", legend.direction = "horizontal")
> ggplot(pima,aes(x=diastolic,y=diabetes)) + geom_point(size=1) +
    facet_grid(~ test)
```

Notice how the first plot, shown in Figure 1.4, has a third element. A *theme* specifying an option regarding the appearance of the plot. In this case, we specify where the legend should appear. The second plot uses two panels. Sometimes this is the better option when crowded plots make different symbols hard to distinguish. I have also reduced the point size in the second plot to make this easier. Full details can be seen in Wickham (2009).

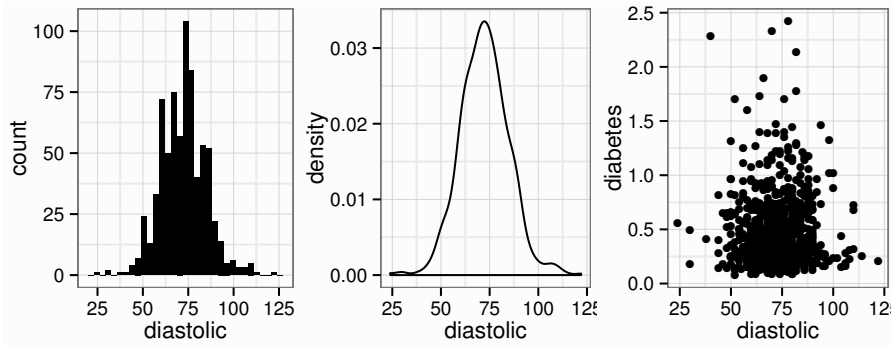



Figure 1.3 Plots of the pima data using ggplot2.

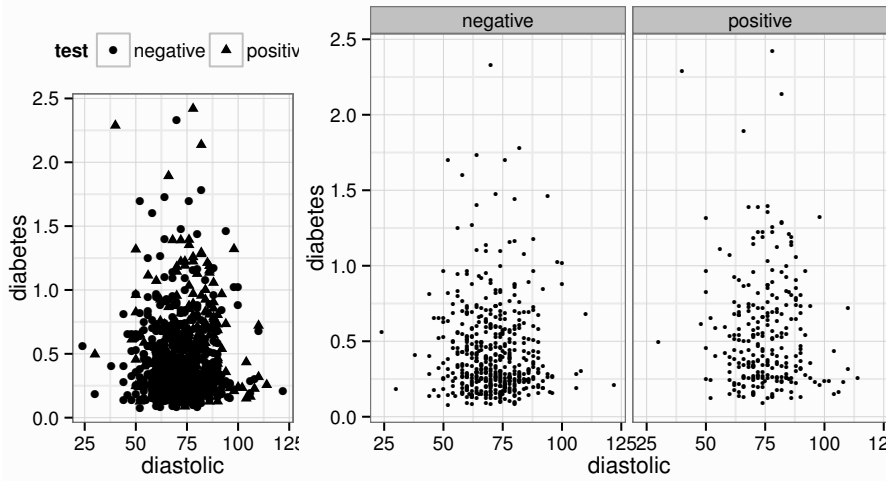



Figure 1.4 Two ways of a distinguishing a factor variable in a bivariate scatter plot.

Good graphics are vital in data analysis. They help you avoid mistakes and suggest the form of the modeling to come. They are also important in communicating your analysis to others. Many in your audience or readership will focus on the graphs. This is your best opportunity to get your message over clearly and without misunderstanding. In some cases, the graphics can be so convincing that the formal analysis becomes just a confirmation of what has already been seen.

## 1.3 When to Use Linear Modeling

Linear modeling is used for explaining or modeling the relationship between a single variable $Y$, called the *response*, *outcome*, *output* or *dependent* variable; and one or more *predictor*, *input*, *independent* or *explanatory* variables, $X_1, \ldots, X_p$, where $p$ is the number of predictors. We recommend you avoid using the words *independent*

and *dependent* variables for $X$ and $Y$ as these are easily confused with the broader meanings of terms. *Regression analysis* is another term used for linear modeling although regressions can also be nonlinear.

When $p = 1$, it is called *simple* regression but when $p > 1$ it is called *multiple* regression or sometimes *multivariate* regression. When there is more than one response, then it is called *multivariate multiple* regression or sometimes (confusingly) multivariate regression. We will not cover this in this book, although you can just do separate regressions on each $Y$.

The response must be a continuous variable, but the explanatory variables can be continuous, discrete or categorical, although we leave the handling of categorical explanatory variables to later in the book. Taking the example presented above, a regression with `diastolic` and `bmi` as $X$s and `diabetes` as $Y$ would be a multiple regression involving only quantitative variables which we tackle first. A regression with `diastolic` and `test` as $X$s and `bmi` as $Y$ would have one predictor that is quantitative and one that is qualitative, which we will consider later in Chapter 14 on *analysis of covariance*. A regression with `test` as $X$ and `diastolic` as $Y$ involves just qualitative predictors — a topic called *analysis of variance* (*ANOVA*), although this would just be a simple two sample situation. A regression of `test` as $Y$ on `diastolic` and `bmi` as predictors would involve a qualitative response. A *logistic regression* could be used, but this will not be covered in this book.

Regression analyses have two main objectives:

1. Prediction of future or unseen responses given specified values of the predictors.
2. Assessment of the effect of, or relationship between, explanatory variables and the response. We would like to infer causal relationships if possible.

You should be clear on the objective for the given data because some aspects of the resulting analysis may differ. Regression modeling can also be used in a descriptive manner to summarize the relationships between the variables. However, most end users of data have more specific questions in mind and want to direct the analysis toward a particular set of goals.

It is rare, except in a few cases in the precise physical sciences, to know (or even suspect) the true model. In most applications, the model is an empirical construct designed to answer questions about prediction or causation. It is usually not helpful to think of regression analysis as the search for some true model. The model is a means to an end, not an end in itself.

## 1.4 History

In the $18^{th}$ century, accurate navigation was a difficult problem of commercial and military interest. Although, it is relatively easy to determine latitude from Polaris, also known as the North Star, finding longitude then was difficult. Various attempts were made to devise a method using astronomy. Contrary to popular supposition, the Moon does not always show the same face and moves such that about 60% of its surface is visible at some time.

Tobias Mayer collected data on the locations of various landmarks on Moon,

including the Manilius crater, as they moved relative to the earth. He derived an equation describing the motion of the moon (called *libration*) taking the form:

$$\texttt{arc} = \beta + \alpha\texttt{sinang} + \gamma\texttt{cosang}$$

He wished to obtain values for the three unknowns $\alpha, \beta$ and $\gamma$. The variables `arc`, `sinang` and `cosang` can be observed using a telescope. A full explanation of the story behind the data and the derivation of the equation can be found in Stigler (1986).

Since there are three unknowns, we need only three distinct observations of the set of three variables to find a unique solution for $\alpha, \beta$ and $\gamma$. Embarassingly for Mayer, there were 27 sets of observations available. Astronomical measurements were naturally subject to some variation and so there was no solution that fit all 27 observations. Let's take a look at the first few lines of the data:

```
> data(manilius, package="faraway")
> head(manilius)
    arc sinang  cosang group
1 13.167 0.8836 -0.4682     1
2 13.133 0.9996 -0.0282     1
3 13.200 0.9899  0.1421     1
4 14.250 0.2221  0.9750     3
5 14.700 0.0006  1.0000     3
6 13.017 0.9308 -0.3654     1
```

Mayer's solution was to divide the data into three groups so that observations within each group were similar in some respect. He then computed the sum of the variables within each group. We can also do this:

```
> (moon3 <- aggregate(manilius[,1:3],list(manilius$group), sum))
  Group.1    arc  sinang  cosang
1       1 118.13  8.4987 -0.7932
2       2 140.28 -6.1404  1.7443
3       3 127.53  2.9777  7.9649
```

Now there are just three equations in three unknowns to be solved. The solution is:

```
> solve(cbind(9,moon3$sinang,moon3$cosang), moon3$arc)
[1] 14.54459 -1.48982  0.13413
```

Hence the computed values of $\alpha, \beta$ and $\gamma$ are -1.49, 14.5 and 0.134 respectively. One might question how Mayer selected his three groups, but this solution does not seem unreasonable.

Similar problems with more linear equations than unknowns continued to arise until 1805, when Adrien Marie Legendre published the method of least squares. Suppose we recognize that the equation is not exact and introduce an error term, $\varepsilon$:

$$\texttt{arc}_i = \beta + \alpha\texttt{sinang}_i + \gamma\texttt{cosang}_i + \varepsilon_i$$

where $i = 1, \ldots, 27$. Now we find $\alpha, \beta$ and $\gamma$ that minimize the sum of the squared errors: $\sum \varepsilon^2$. We will investigate this in much greater detail in the chapter to follow but for now we simply present the solution using R:

```
> lmod <- lm(arc ~ sinang + cosang, manilius)
> coef(lmod)
(Intercept)      sinang      cosang
  14.561624   -1.504581    0.091365
```

We observe that this solution is quite similar to Mayer's. The least squares solution is more satisfactory in that it requires no arbitrary division into groups. Carl Friedrich Gauss claimed to have devised the method of least squares earlier but without publishing it. At any rate, he did publish in 1809 showing that the method of least squares was, in some sense, optimal.

For many years, the method of least squares was confined to the physical sciences where it was used to resolve problems of overdetermined linear equations. The equations were derived from theory and least squares was used as a method to fit data to these equations to estimate coefficients like $\alpha, \beta$ and $\gamma$ above. It was not until later in the 19th century that linear equations (or models) were suggested empirically from the data rather than from theories of physical science. This opened up the field to the social and life sciences.

Francis Galton, a nephew of Charles Darwin, was important in this extension of statistics into social science. He coined the term *regression to mediocrity* in 1875 from which the rather peculiar term *regression* derives. Let's see how this terminology arose by looking at one of the datasets he collected at the time on the heights of parents and children in Galton (1886). We load the `HistData` package of historical statistical datasets and plot some of the data as seen in Figure 1.5. You will need to install this package using `install.packages("HistData")` if you have not already done so.

```
> data(GaltonFamilies, package="HistData")
> plot(childHeight ~ midparentHeight, GaltonFamilies)
```

We see that `midparentHeight`, defined as the father's height plus 1.08 times the mother's height divided by two, is correlated with the `childHeight`, both in inches. Now we might propose a linear relationship between the two of the form:

$$\texttt{childHeight} = \alpha + \beta \texttt{midparentHeight} + \varepsilon$$

We can estimate $\alpha$ and $\beta$ using R and plot the resulting fit as follows:

```
> lmod <- lm(childHeight ~ midparentHeight, GaltonFamilies)
> coef(lmod)
    (Intercept) midparentHeight
       22.63624         0.63736
> abline(lmod)
```

For the simple case of a response $y$ and a single predictor $x$, we can write the equation in the form:

$$\frac{y-\bar{y}}{SD_y} = r\frac{(x-\bar{x})}{SD_x}$$

where $r$ is the correlation between $x$ and $y$. The equation can be expressed in words as: the response in standard units is the correlation times the predictor in standard units. We can verify that this produces the same results as above by rearranging the equation in the form $y = \alpha + \beta x$ and computing the estimates:

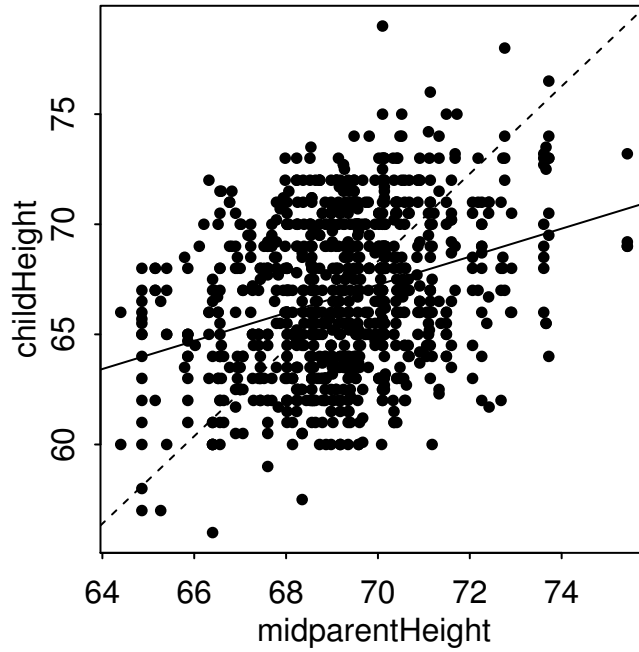



Figure 1.5 The height of child is plotted against a combined parental height defined as (father's height + 1.08 × mother's height)/2.

```
> (beta <- with(GaltonFamilies, cor(midparentHeight, childHeight) * sd
    (childHeight) / sd(midparentHeight)))
[1] 0.63736
> (alpha <- with(GaltonFamilies, mean(childHeight) - beta * mean(
    midparentHeight)))
[1] 22.636
```

Now one might naively expect that a child with parents who are, for example, one standard deviation above average in height, to also be one standard deviation above average in height, give or take. The supposition would set $r = 1$ in the equation and leads to a line which we compute and plot below:

```
> (beta1 <- with(GaltonFamilies, sd(childHeight) / sd(midparentHeight)
    ))
[1] 1.9859
> (alpha1 <- with(GaltonFamilies, mean(childHeight) - beta1 * mean(
    midparentHeight)))
[1] -70.689
> abline(alpha1, beta1, lty=2)
```

The resulting dashed line is added to Figure 1.5. The lines cross at the point of the averages. We can see that a child of tall parents is predicted by the least squares line to have a height which is above average but not quite as tall as the parents as the dashed line would have you believe. Similarly children of below average height

parents are predicted to have a height which is still below average but not quite as short as the parents. This is why Galton used the phrase "regression to mediocrity" and the phenomenom is sometimes called the regression effect.

This applies to any $(x,y)$ situation like this. For example, in sports, an athlete may have a spectacular first season only to do not quite as well in the second season. Sports writers come up with all kinds of explanations for this but the regression effect is likely to be the unexciting cause. In the parents and children example, although it does predict that successive descendants in the family will come closer to the mean, it does not imply the same of the population in general since random fluctuations will maintain the variation, so no need to get too pessimistic about mediocrity! In many other applications of linear modeling, the regression effect is not of interest because different types of variables are measured. Unfortunately, we are now stuck with the rather gloomy word of regression thanks to Galton.

Regression methodology developed rapidly with the advent of high-speed computing. Just fitting a regression model used to require extensive hand calculation. As computing hardware has improved, the scope for analysis has widened. This has led to an extensive development in the methodology and the scale of problems that can be tackled.

## Exercises

1. The dataset `teengamb` concerns a study of teenage gambling in Britain. Make a numerical and graphical summary of the data, commenting on any features that you find interesting. Limit the output you present to a quantity that a busy reader would find sufficient to get a basic understanding of the data.
2. The dataset `uswages` is drawn as a sample from the Current Population Survey in 1988. Make a numerical and graphical summary of the data as in the previous question.
3. The dataset `prostate` is from a study on 97 men with prostate cancer who were due to receive a radical prostatectomy. Make a numerical and graphical summary of the data as in the first question.
4. The dataset `sat` comes from a study entitled "Getting What You Pay For: The Debate Over Equity in Public School Expenditures." Make a numerical and graphical summary of the data as in the first question.
5. The dataset `divusa` contains data on divorces in the United States from 1920 to 1996. Make a numerical and graphical summary of the data as in the first question.